Instagram Performance Analytics for Sugar Cosmetics

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Tool: Microsoft Power BI | Digital Marketing and Business Intelligence project

1. Project overview

Sugar Cosmetics is an Indian beauty brand with an active Instagram presence. In this project I acted as a social media analyst. I was given a dataset of the brand's Instagram posts and asked to build an interactive dashboard that shows how the account performs, then use it to recommend what the brand should do differently.

The project brief lists 10 tasks: a KPI overview, a monthly posting trend, the top 10 posts by likes and by engagement rate, a post type comparison, a day-of-week analysis, an engagement funnel, a hashtag analysis, a final interactive dashboard, and written strategic recommendations. The brief mentions Tableau, but my mentor approved Power BI, so the whole project was built in Power BI.

2. Objectives

- Measure the overall health of the account (reach, likes, comments, follows and engagement rate).
- Find which formats, days, hashtags, campaigns and sources drive the most engagement.
- Show where viewers drop off between seeing a post and interacting with it.
- Combine everything into one interactive dashboard.
- Turn the findings into practical recommendations for the brand.

3. The dataset

The dataset has 50,000 Instagram posts and 14 columns, covering April 2024 to April 2025. It had no missing values, no duplicate rows and no messy text, so little cleaning was needed.

Columns: Post Date, Post Type, Campaign Type, Hashtags, Audience Emotion, Engagement Source, Impressions, Reach, Likes, Comments, Shares, Saves, Profile Visits and Follows.

The dataset differs from the brief in a few ways:

- The post types are Image, Video, Reel and Story. There is no Carousel.
- There is no caption column, so posts are identified by Post ID, type, campaign and date.
- It has extra columns (campaign type, audience emotion, engagement source, profile visits and follows), which I used for deeper insights.
- April 2024 and April 2025 are partial months, so their dips in the trend chart are not real declines.

4. Method

Data preparation in Power Query

- Set the correct data type for every column (dates as Date, counts as Whole Number, labels as Text).
- Added a PostID column, numbered from 1 to 50,000.
- Added helper columns: Month Start, Month Year, Weekday and Weekday Number, so charts sort in the right order.
- Split the Hashtags column (up to three tags per post) into a separate Hashtags table, one tag per row, linked to Posts by PostID. Keeping it separate stops totals from being inflated.

Data model

The model has a Posts table and a Hashtags table joined one-to-many on PostID, with the filter direction set to both so a hashtag selection filters the posts. A small helper table, Funnel Stages, lists the six funnel stages for the funnel chart.

Measures (DAX)

- Total Posts, Total Likes, Total Comments, Total Shares, Total Saves, Total Reach, Total Impressions, Total Profile Visits and Total Follows are sums or counts of the matching column.
- Total Engagement is likes plus comments plus shares plus saves.
- Engagement Rate is Total Engagement divided by Total Reach.

Report design

The report has 8 pages at 1920 x 1080: KPI Overview, Top Posts, Post Type Analysis, Day of Week Analysis, Engagement Funnel, Hashtag Analysis, Campaign and Source Analysis, and the final Dashboard. I applied a custom dark wine and pink theme, and every chart carries a subtitle that states its main finding.

5. Findings

Overall performance (Tasks 1 and 2)

The account has 50,000 posts, 987.4 million reach, 1.28 billion impressions, 147.7 million likes, 18.5 million comments, 8.1 million shares, 8.9 million saves and 8.87 million follows. The overall engagement rate is 18.55%. The brand posts steadily, with about 3,900 to 4,300 posts in every full month.

Top posts (Tasks 3 and 4)

The top 10 posts by likes have 11,043 to 11,392 likes, all with reach above 44,000. The top 10 by engagement rate are different posts, with 33.8% to 34.1% engagement, but some have only 1,791 to 3,996 likes. These hidden gems show that likes alone do not measure quality.

Post type (Task 5)

All four formats perform within about 1% of each other. Engagement rate is 18.63% for Story, 18.60% for Reel, 18.56% for Video and 18.42% for Image. Average likes range only from 2,941 to 2,964. Image has the highest average reach (19,864) but the lowest engagement rate.

Day of week (Task 6)

Friday has the highest engagement rate (18.61%), followed by Saturday (18.59%). Sunday is lowest (18.47%). The gap is only 0.14 points. Posting is balanced, from 7,028 posts on Sunday to 7,355 on Tuesday. Wednesday has the highest average likes (2,974) and Saturday the lowest (2,926).

Engagement funnel (Task 7)

Of 1,276 million impressions, 77.4% become unique reach (987 million). Likes are 11.6% of impressions, which is only about 15% of the people reached. Comments are 1.45% of impressions, and shares (0.64%) and saves (0.69%) are the weakest stages. In addition, profile visits are 2.3% of impressions, and about 30% of profile visits end in a follow. The biggest leak is after reach.

Hashtags (Task 8)

#VeganBeauty has the highest engagement rate (18.66%) and the brand's own tag, #SugarCosmetics, the lowest (18.51%). #CrueltyFree has the most posts (20,197) and the biggest reach (401 million). Posts with two or three hashtags average 2,972 to 2,981 likes, against 2,910 for posts with one hashtag. Because a post can carry up to three tags, hashtag counts overlap and should not be added together.

Campaigns, sources and emotions (supports Task 10)

- Campaign type: Influencer Collab is highest (18.63%), then Tutorial (18.59%), Giveaway (18.56%) and Product Launch (18.52%). Festive Offer is lowest (18.45%).
- Engagement source: Influencer is highest (18.65%), then Collab (18.63%), Paid (18.55%) and Organic (18.37%). Organic posts have the highest average reach (19,970) but the lowest engagement. Paid posts have the lowest average reach (19,617).
- Audience emotion: Curious is highest (18.66%). The other four sit at 18.52% to 18.53%.

The most important pattern is that individual posts range from about 5.6% to 34.1% engagement, but no category explains that spread. What makes a post perform is mostly its content and creative, which this dataset does not capture.

6. The dashboard (Task 9)

The final dashboard combines six KPI cards (posts, engagement rate, follows, reach, likes and comments) and six charts (monthly trend, funnel, hashtags, day of week, campaign type and engagement source). Three dropdown slicers (Post Type, Month Year and Hashtag) filter the whole page, and clicking any bar cross-filters every other visual, which works like filter actions in Tableau. I checked the numbers with the slicers against the raw data, for example Reel gives 12,599 posts and an 18.60% engagement rate.

7. Strategic recommendations (Task 10)

When to post

Use Friday and Saturday for key launches, since they have a slight edge, but keep posting every day because the gap is small.

What to post

- Do more influencer collaborations, since influencer-led and collab posts beat organic.
- Use Story and Reel formats a little more, and be careful with Image.
- Use curious, question-led hooks.
- Rework the Festive Offer creative, which is the weakest campaign.

How to grow engagement and followers

- Add clear calls to action to save and share, because shares and saves are under 0.7% of impressions.
- Ask questions in captions to lift comments, which are only 12.5% of the number of likes.
- Use two to three values-based hashtags such as #VeganBeauty and #CrueltyFree, and treat the brand tag as a supporting tag.
- Improve the profile (bio, highlights and pinned posts) to lift the roughly 30% profile-visit-to-follow rate.

Gaps in the current approach

- Organic content has the widest reach but the weakest engagement.
- Paid posts do not clearly outperform organic ones.
- Shares and saves are very low compared with likes.
- The dataset has no captions, creative or follower baseline, so the team cannot tell which content ideas work.

Campaign ideas to test

1. An influencer tutorial series, measured on saves and shares.
2. A Friday versus Wednesday launch of similar products.
3. Values-based hashtags against the brand tag.
4. A Festive Offer refresh with a giveaway angle.
5. A month of posts with a "save for later" call to action.

8. Limitations

- Most differences between categories are under one percentage point, so the recommendations should be treated as tests, not certain strategy shifts.
- April 2024 and April 2025 are partial months.
- Hashtag totals overlap, because one post can count under up to three hashtags.
- The dataset has no captions, creative details or follower baseline.

9. Conclusion

Sugar Cosmetics posts consistently and earns strong reach, with an overall engagement rate of 18.55%. Format, weekday and hashtag choices make only small differences. The clearest gaps are that influencer-led content beats organic, and that the funnel leaks after reach, with only about 15% of the people reached liking a post and under 0.7% of impressions turning into shares or saves. The brand should lean on influencer collaborations, build save-worthy and shareable content, and run structured tests to find out which creative ideas really drive engagement.